Supplemental Material
Self-organization in bi-motility mixture of model microswimmers

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I. Movie Captions

MOVIE 1. The movie corresponds to the simulation of system under circular confinement with radius $R = 100$ and $N_a = N_b = 75$. The green colored swimmers are faster than the red colored swimmers. ν_a(green)= 0.0065 and ν_b(red)= 0.0035. The movie shows how swimmers of both times partly mix together even after getting segregated. The arrows in the movie point towards the region where the clusters of slow swimmers near to the wall are fragmented by the fast swimmers and thus the slower ones move towards the center.

MOVIE 2. The movie corresponds to the simulation of system under circular confinement with radius $R = 100$ and $N_a = N_b = 75$. The green colored swimmers are faster than the red colored swimmers. ν_a(green)= 0.00505 and ν_b(red)= 0.00495. The arrows in the movie point towards the mixed clusters in which the greater number of slower ones can be seen at the back utilizing the thrust of faster swimmers. After some time, the slower ones segregate out of the cluster.

II. Figures

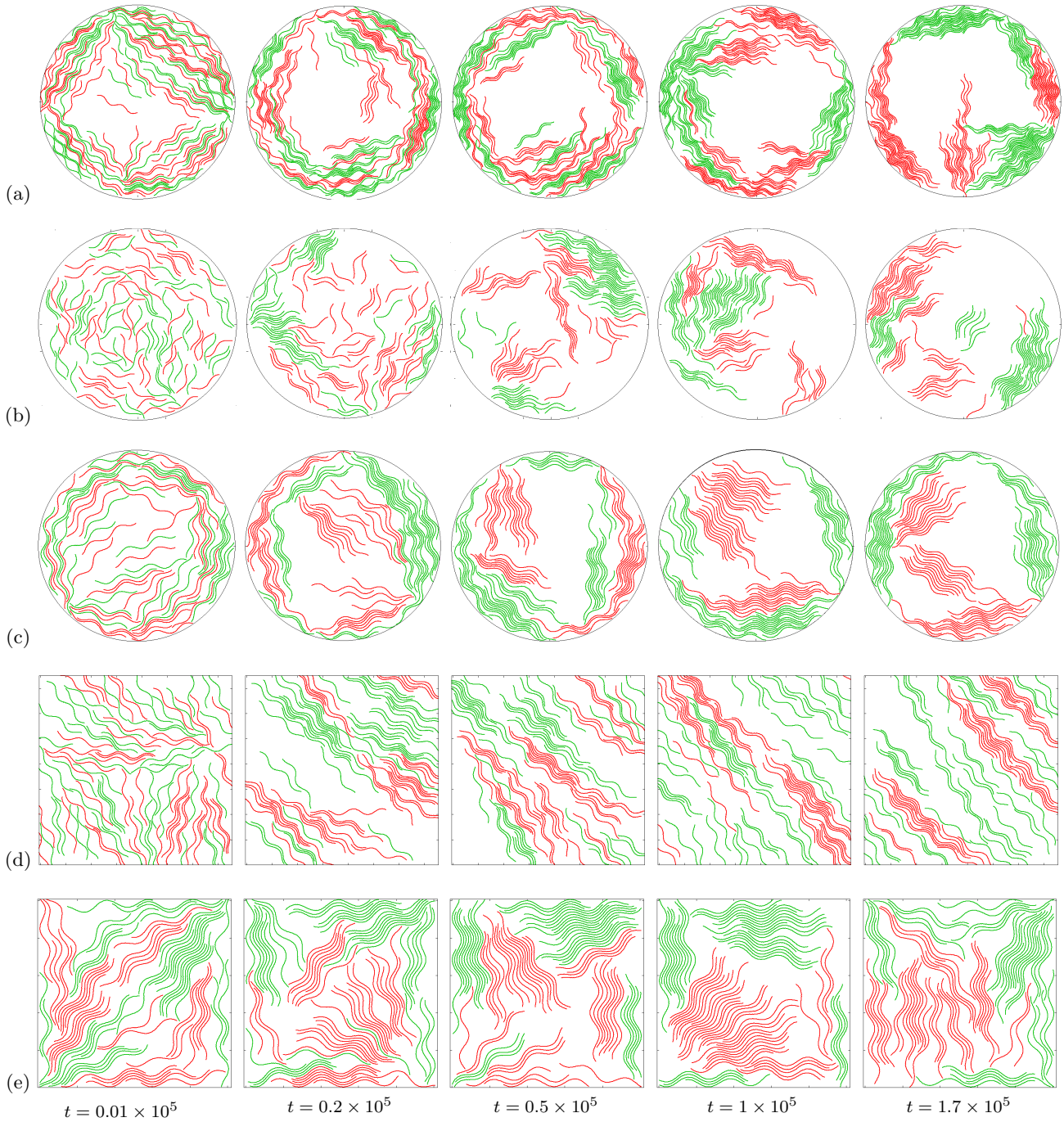


FIG. S1. Snapshots showing time evolution of the mixture for different system parameters. Here green colored swimmers are faster than the red ones such that $\delta\nu = 0.8$. (a) Standard system under circular boundary with $R = 100$. (b) system under circular boundary with $R = 80$ and $L_c = 25$, i.e. half of that in other systems ($L_c = 50$). (c) system under circular boundary with $R = 80$ and amplitude, $A = 2.94$, while in all other systems, $A = 2.27$. (d) has periodic boundary of 150×150. (e) is under square confinement with $L = 100$. Number density $\rho = 0.0048$ in each. All the systems segregate as time evolves.

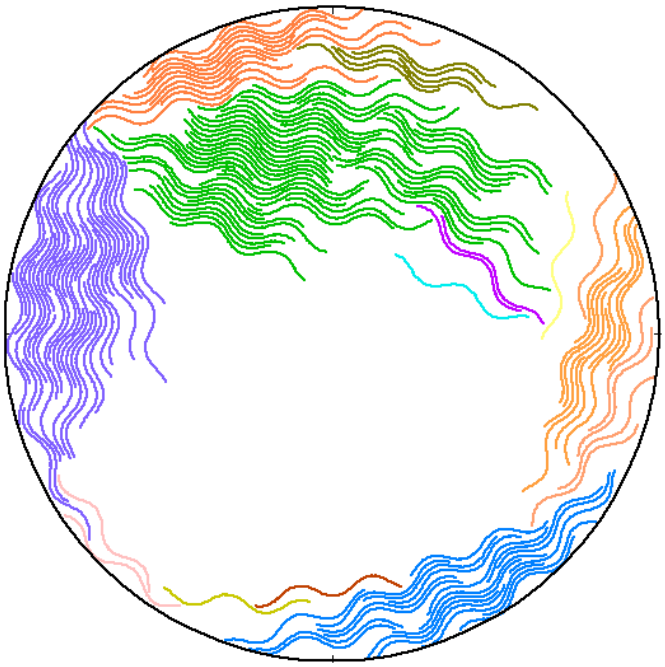
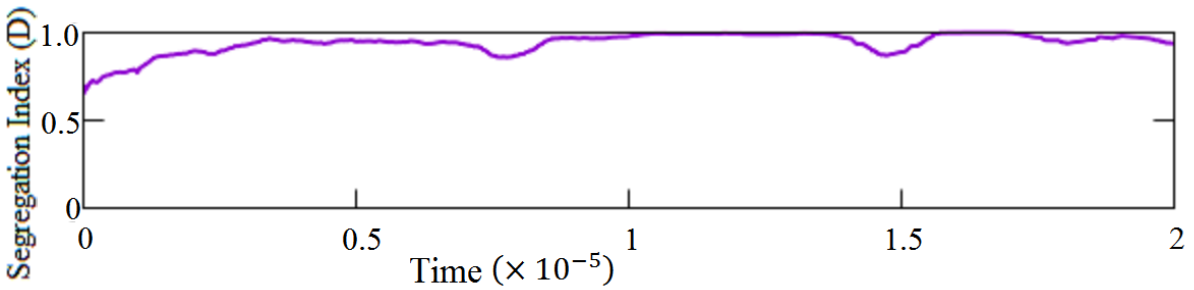
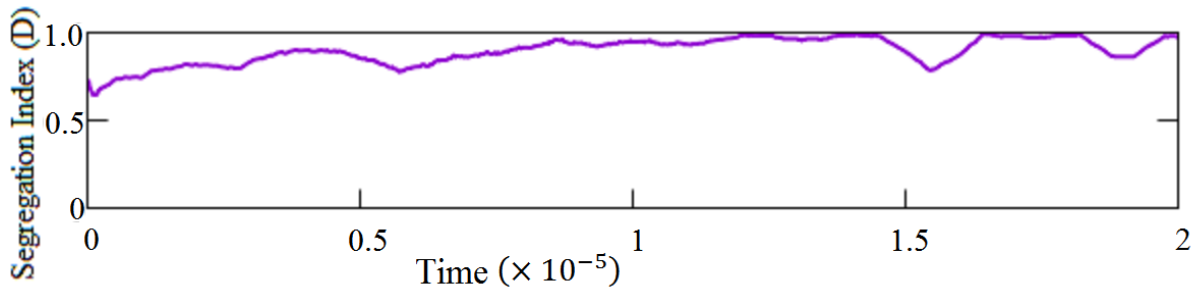


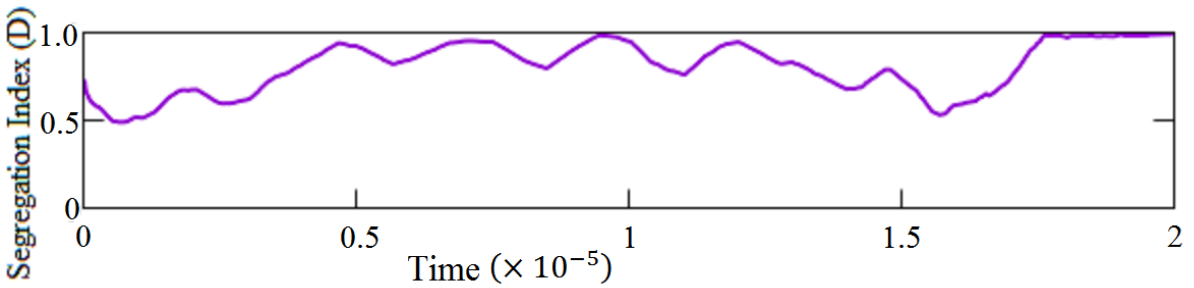
FIG. S2. Illustration showing different clusters as per our definition. Each color represents a different cluster. The snapshot is of a system with 150 swimmers in circular confinement of radius 100. The system contains homogeneous mixture of swimmers with relative difference in beating frequencies ($\delta\nu = 0.1$), $\nu_a = 0.00525$ and $\nu_b = 0.00475$. The snapshot is of the same state shown in Fig. 1a.



(a)



(b)



(c)

FIG. S3. Segregation Index, D vs time for three systems having 150 swimmers uniformly distributed at $t = 0$ under circular confinement with $R = 100$. The systems differ in the beating frequency of swimmers, $\delta\nu_a = 0.6$, $\delta\nu_b = 0.3$ and $\delta\nu_c = 0.06$ and the average frequency in all cases is 0.005. The calculations are done for the same systems studied in Fig. 4 and in all the cases the system finally attains a segregation index close to 1 i.e. each system, regardless of $\delta\nu$ reaches a completely segregated state.